

Question:

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient’s X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient’s condition?

Answer Choices:

- A. It activates TLR5 on the surface of macrophages.
- B. It primarily induces the Th1-cell response.
- C. Peptidoglycan is its major antigen that induces an immune response.
- D. This pathogen evades the immune response by encapsulation.

Reasoning + Answer:

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- The patient has community-acquired pneumonia and is responding to cephalexin, which is a penicillin antibiotic. This suggests a bacterial pathogen.

- Cephalexin targets gram-positive bacteria, which means the pathogen is likely a gram-positive bacteria.

- Gram-positive bacteria have peptidoglycan in their cell wall, which can induce immune responses.

- The immune response to bacteria often involves both Th1 and Th17 cell responses, with Th1 dominating.

- However, the question specifically asks for the best description, and the most common and primary response is often Th1.

- Peptidoglycan in gram-positive bacteria is a key component that stimulates the immune system.

- Encapsulation by pathogens is more associated with fungal pathogens and some bacteria like Streptococcus pneumoniae, which is less likely given the response to penicillin.

Therefore, the best answer among the choices is: It primarily induces the Th1-cell response. While peptidoglycan is an important antigen, the primary immune response to most bacteria is Th1, and the question focuses on the best overall description.

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<answer>B. It primarily induces the Th1-cell response.</answer>